Lecture 02: Data Modeling
(Requirement Analysis, Business Rules and Constraints)

ITCS216 Fundamentals of Database Systems
Semester 1/2026

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This lecture is adopted from ITCS241 content in 1/2022 and collarborate with Asst. Prof. Dr. Preecha Tangworakitthaworn

Lesson Learning Outcomes

After this lesson, students are able to:

- Understand business rules and its importance to database design.
- Extract the business rules from the given requirements.
- Understand the data modeling.

Database Concept

Recommend

Three Schema Architecture

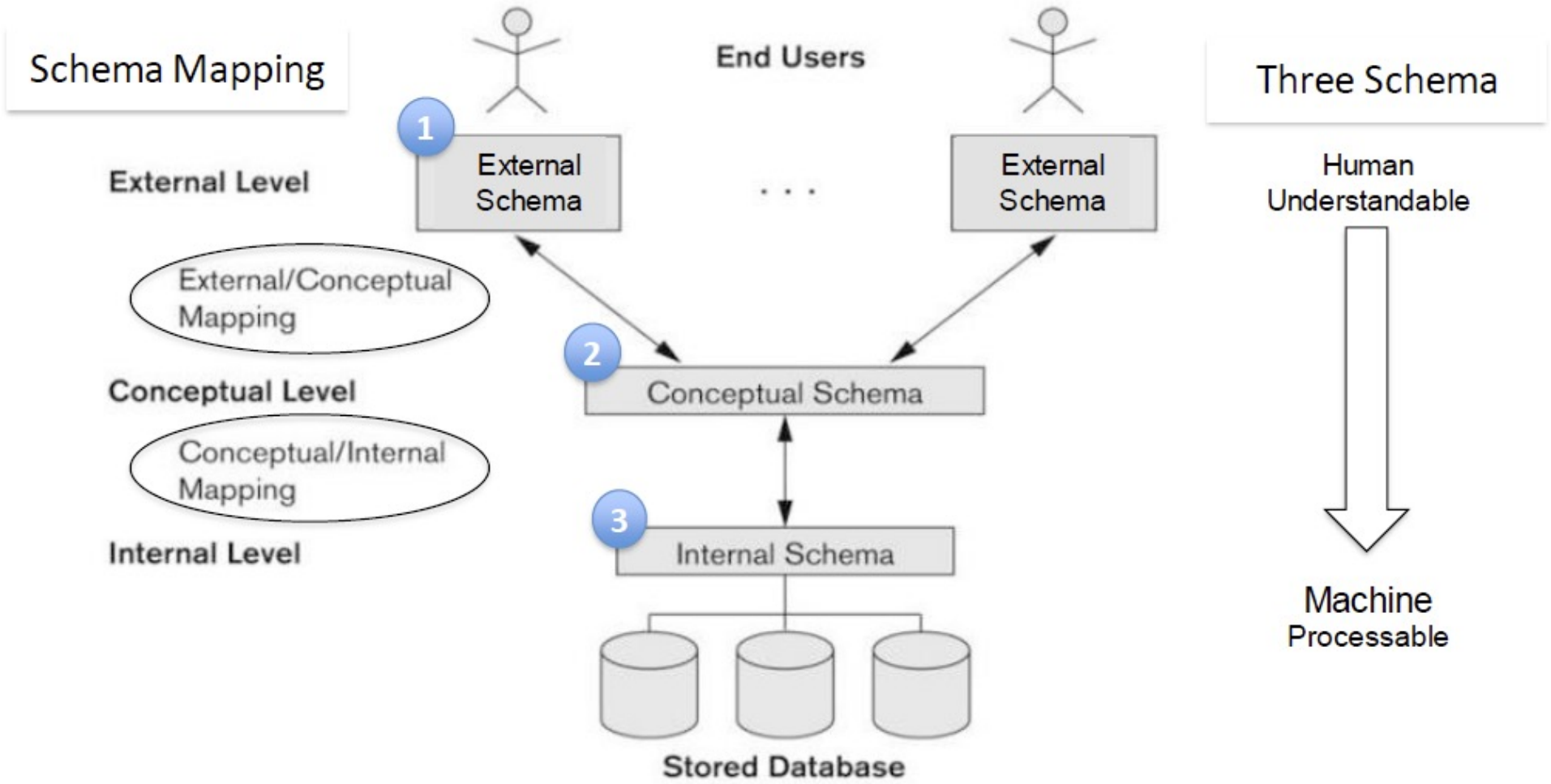


FIGURE 1.1 THE PERVASIVE NATURE OF DATABASES

A Day In Susan's Life

See how many databases she interacts with each day

Before leaving for work, Susan checks her Facebook and Twitter accounts



Where is the data about the friends and groups stored?

Where are the "likes" stored and what would they be used for?



On her lunch break, she picks up her prescription at the pharmacy



Where is the pharmacy inventory data stored?

What data about each product will be in the inventory data?

What data is kept about each customer and where is it stored?



After work, Susan goes to the grocery store



Where is the product data stored?

Is the product quantity in stock updated at checkout?

Does she pay with a credit card?



At night, she plans for a trip and buys airline tickets and hotel reservations online



Where does the online travel website get the airline and hotel data from?

What customer data would be kept by the website?

Where would the customer data be stored?



Then she makes a few online purchases



Where are the product and stock data stored?

Where does the system get the data to generate product "recommendations" to the customer?

Where would credit card information be stored?



Business Domain

- Business Domain refers to a specific area or sector of business activities, operations, or expertise within an organization, company, or industry.

Business Domain	“Book Store”	“Airline”	“Coffee Shop”
Company	Kinokuniya, ASIA BOOKS, นายอินทร์ naiin.com, SE-ED Book Center, B2S	THAI, Bangkok Airways, CHINA AIRLINES, KOREAN AIR, JAL	Starbucks, Café Amazon, Inthanin

Example: Bookstore

English Books > Computers an

thailand.kinokuniya.com/t/books/english-books/computers-and-internet

BOOKS Kinokuniya THAILAND

Find a Store | Help | Enquiry | ENGLISH | ภาษาไทย | 日本語

All | ADVANCED SEARCH | Welcome! Guest Your Page | Cart

English Books | Chinese Books | Japanese Books | Thai Books | รายชื่อสำนักพิมพ์ | Comics/Manga | Lifestyle | Stationery | More

Top > English Books > Computers and Internet

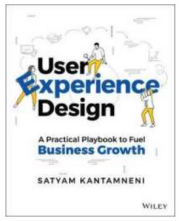
English Books

- Animals and Pets
  - Cats
  - Dogs
  - Other Domestic Pets
- Art and Design
  - Architecture
  - Fashion Design
  - Fine Arts
  - Graphic Design & Product Design
  - Interior Design
  - Photography
- Biography
  - Business
  - Historical & Political
  - True Crime
  - Other Biographies
- Business and Economics
  - Accounting
  - Biographies
  - Business Management
  - Business Writing (Reports/Resumes)
  - Economics
  - Finance and Investment
  - Sales and Marketing
- Children's Books
  - Babies / Toddlers
  - Pre-Teens (Ages 7-12)

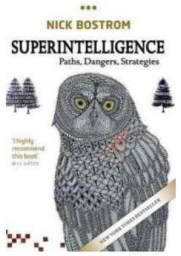
COMPUTERS AND INTERNET

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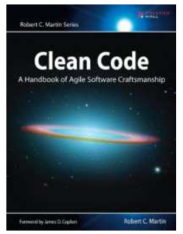
Sort by: Monthly Bestseller | View: 20 | List | Grid | Page 1 of 97



User Experience Design : A Practical Pla... by Kantamneni, Satyam



Superintelligence : Paths, Dangers, Stra... by Bostrom, Nick



Clean Code : A Handbook of Agile Softwar... by Martin,



Clean Architecture : A Craftsman's Guide... by Martin, Robert

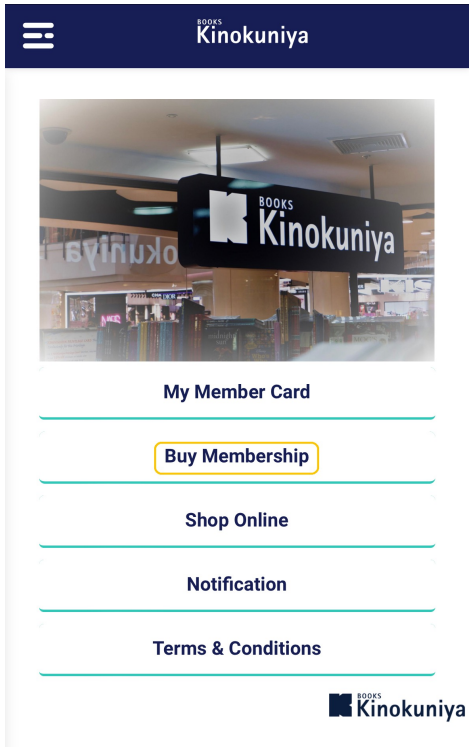


Business process

- **Business process** is a set of structured activities or steps that an organization follows to achieve a specific business goal or objective.
- Business process Involves the systematic flow of tasks, and resources across different stages to produce a desired business outcome.

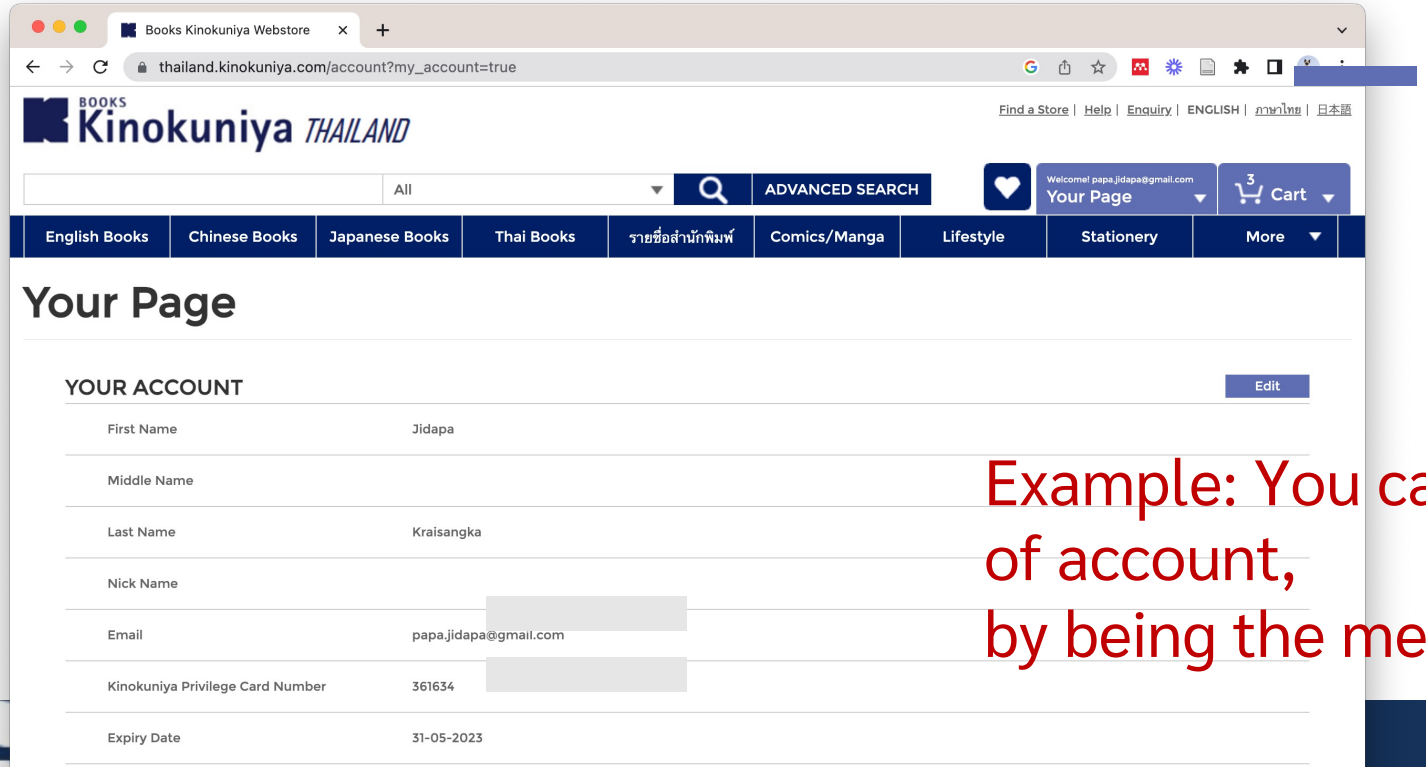
Example In **KNKY** bookstore, the main business process includes:

- Cart Management
- Payment
- Shipment and Delivery
- Membership Account Management



Business Process Analysis

- Analyze the operation within the business process by extracting document examples, or flow of the given operation.
- Interview stakeholders (the ones who are responsible for the operation) for more insight.



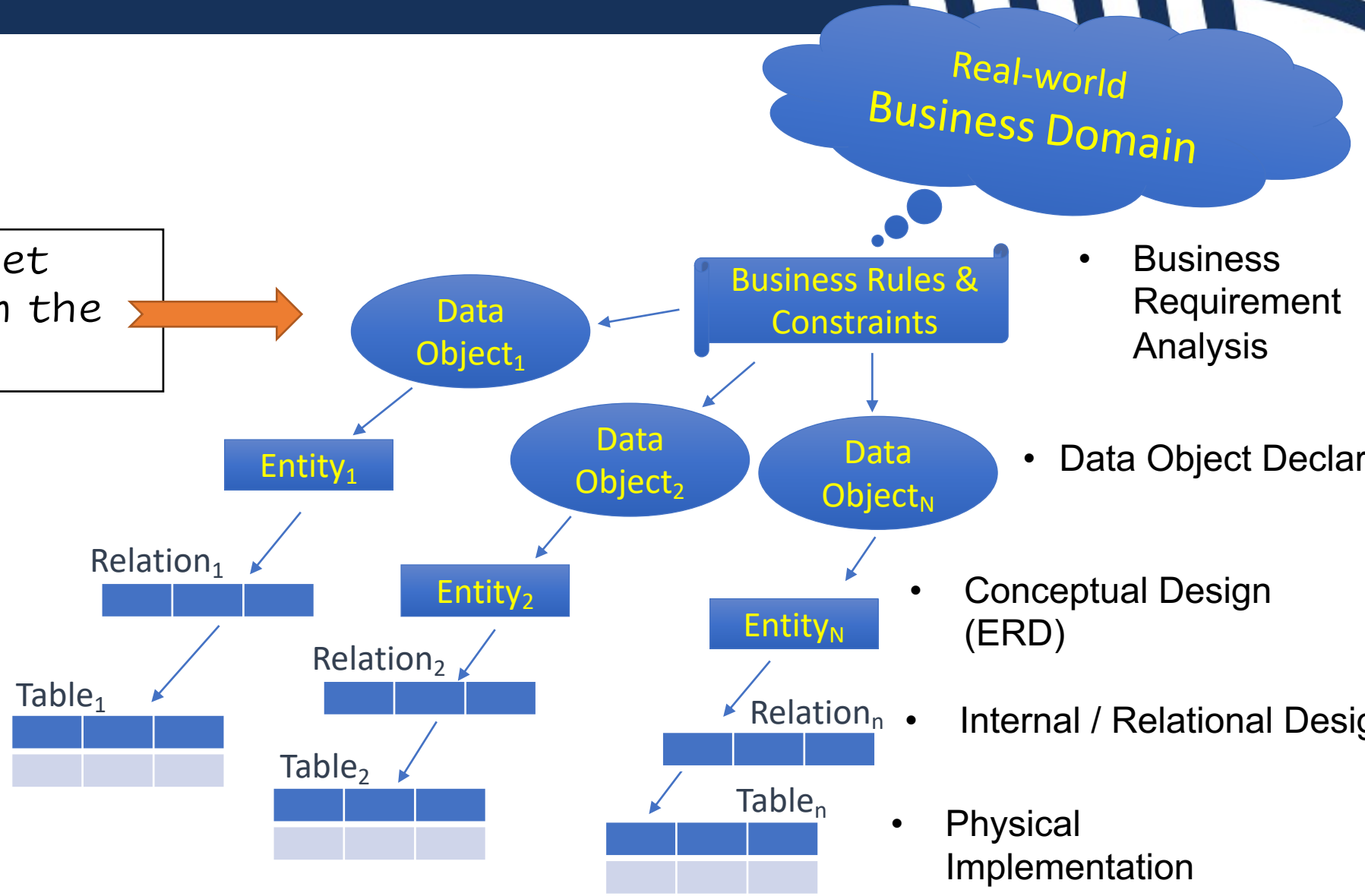
Example: You can study the process of account,
by being the members

Data Object Declaration

- In database design, the Data Object Declaration aims to define the “Data Object Name” and its characteristics, properties, or schema.
- Output of the declaration can support the database design and development process to indicate the DATA OBJECT, then, ENTITY, and rather on to declare the RELATION and the Physical TABLE, in the database system.

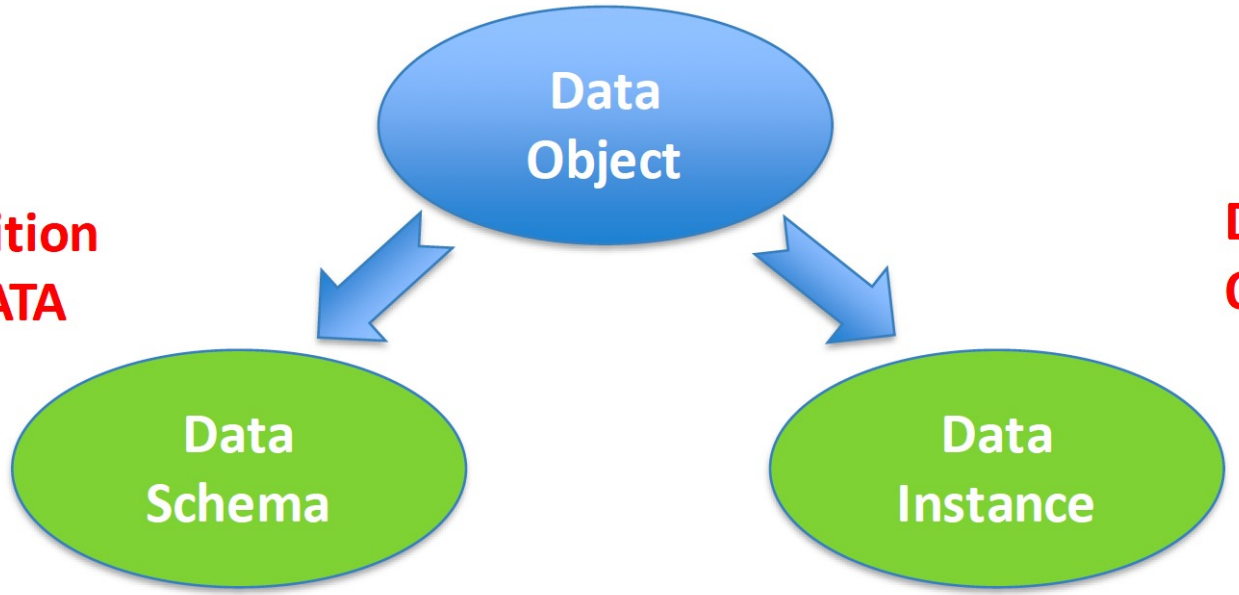
Data Object Declaration

Extract the target data object from the requirements.



- Business Requirement Analysis
- Data Object Declaration
- Conceptual Design (ERD)
- Internal / Relational Design
- Physical Implementation

Nature of Data Object



Data Definition Or METADATA

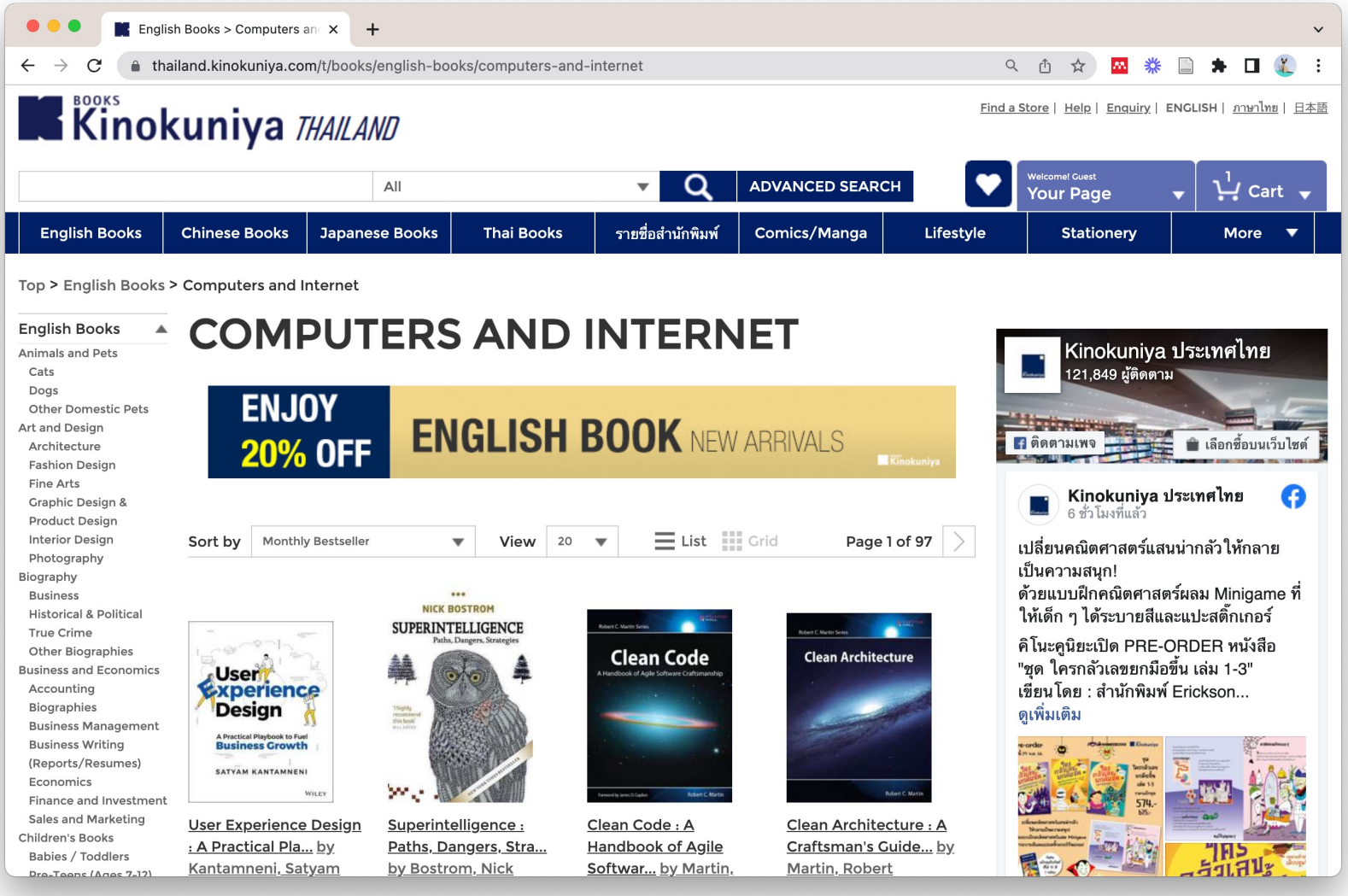
Data Instance Or Data Record

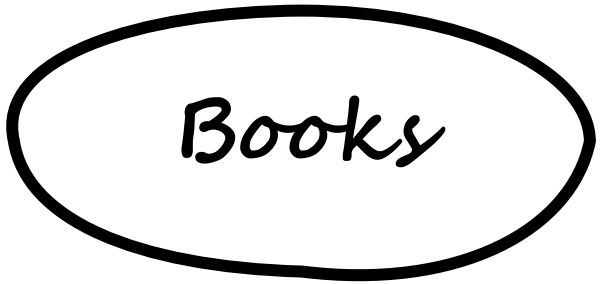
- Skeleton or structure of data
- Properties or Characteristics of data
- Defined as Attribute of data
- For example, Student_Name, Student_ID, Gender

- Raw fact or Raw data
- Structural or Un-structural data
- Defined as Data Record
- For example, Mr.Somboon Sae-tae, 6288000, Male.

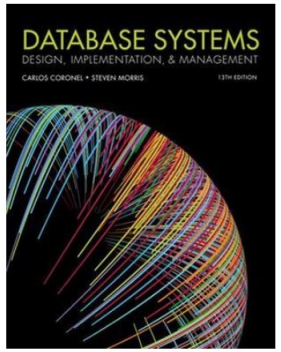
Glossary: MetaData is data about data defined the database structure (or database schema).

Example: Data Object





Example: Data Schema





Database Systems : Design, Implementation, and Management (13th) [Hardcover]

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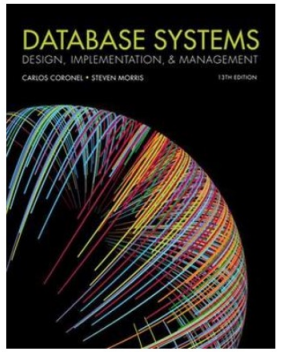
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Example: Data Instance





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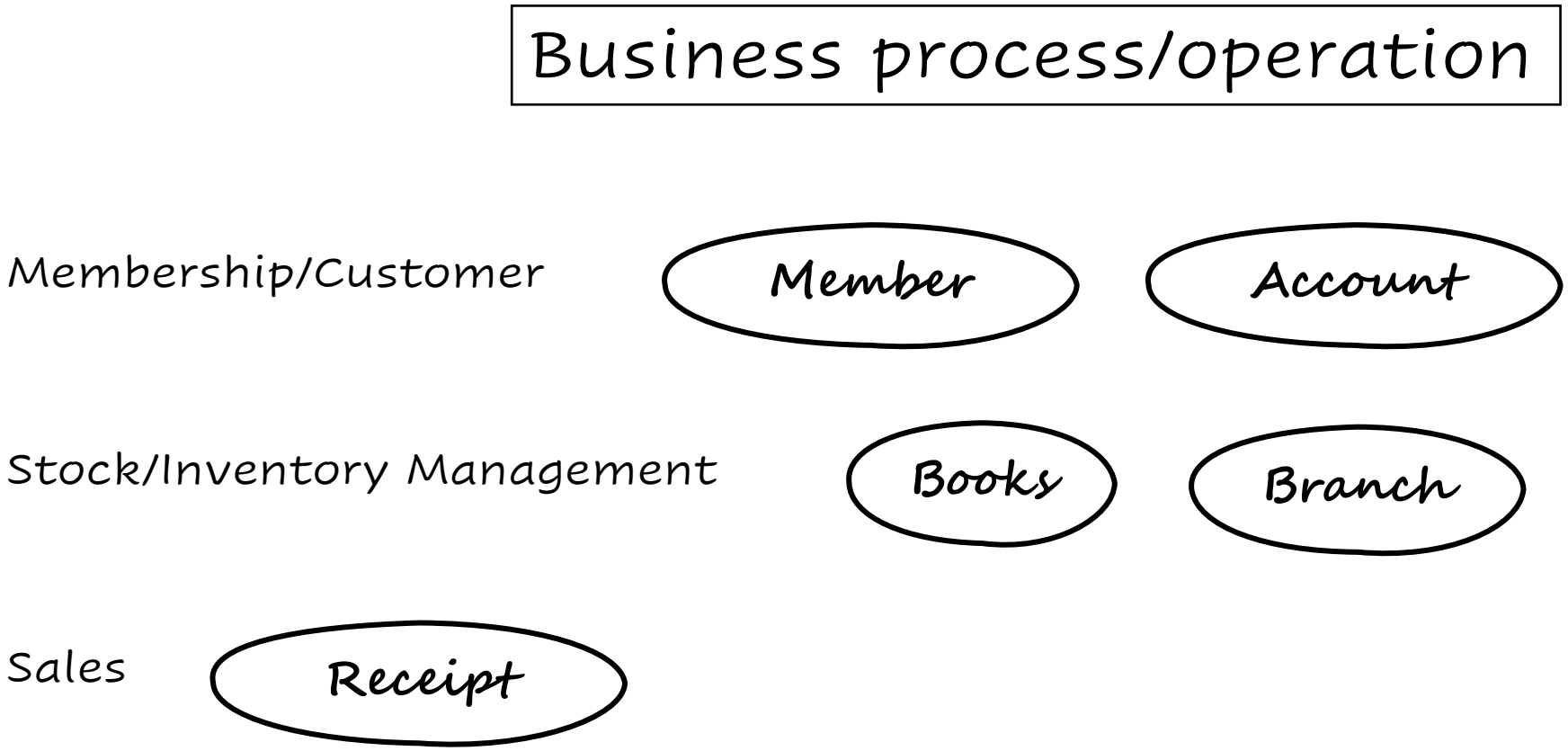


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How we extract the data?

"Understand the business domain / business requirement"



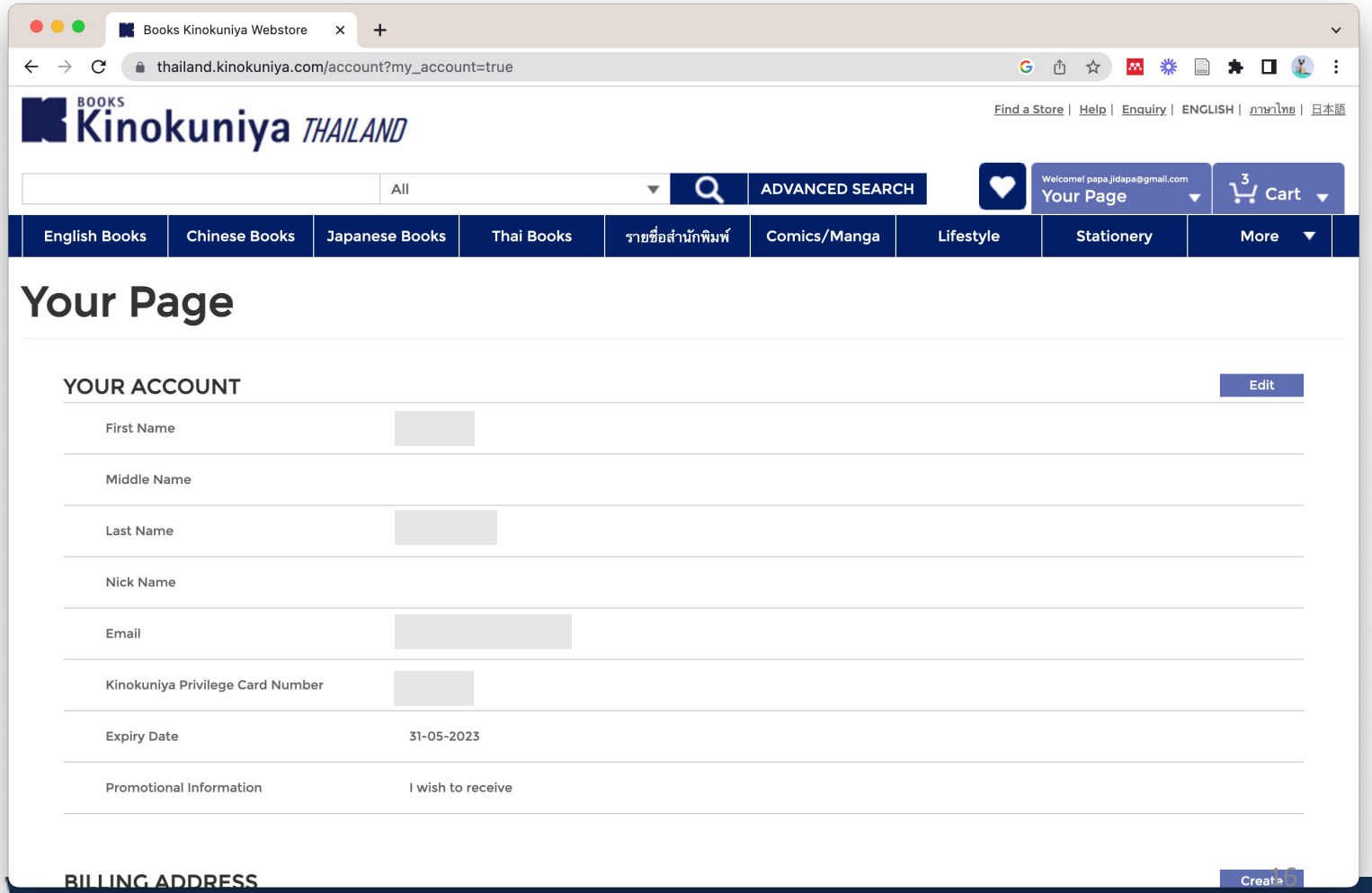
How we extract the data?

"Understand the business domain / business requirement"

Membership/Customer

Member

Account



How do you properly identify entities, attributes, relationships, and constraints?

Business Rules



Business Requirement

The problem:

- Organizations get complicated as they evolve; simply automating the business results in complex, restrictive and inflexible systems.

The Solution:

- Top-down analysis of business requirements represents what's truly required
- Essential: End-users' needs or requirements
- Enterprise: the whole business, or specific business functions/operations.

Business Rules

- A **business rule** is a brief, precise, and unambiguous description of a policy, procedure, or principle within a specific organization.
- Usually, must be rendered in writing and updated to reflect any change in the organization's operational environment.
- Properly written business rules are used to define entities, attributes, relationships, and constraints.

Example:

- A customer may generate many invoices.
- An invoice is generated by only one customer.
- A training session cannot be scheduled for more than 29 employees

Business Rule's Template



- **Template #1**

Data Object Noun	Relationship Verb	Data Object Noun

For example: A Student attends many Classes.

- **Template #2**

Data Object Noun	Container Verb	Data Properties/Attributes Noun

For example: A Student comprises Student ID, Name, and many EMails.

Business Requirement Analysis

Recommend

- Forward analysis

  Analyze on the main object using Active voice.

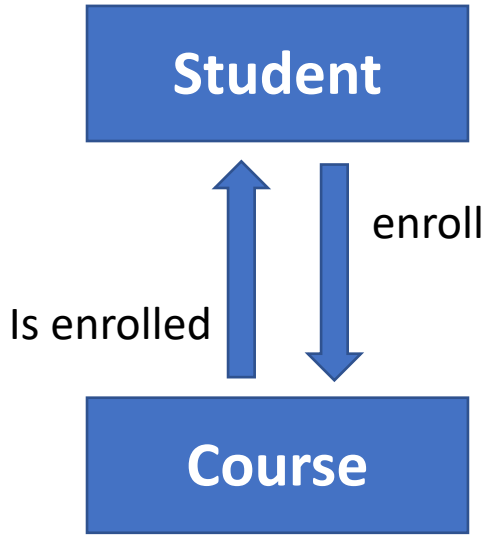
  e.g., A Student enrolls for many courses.

- Backward analysis

  Analyze on the indirect object using Passive voice.

  e.g., A course is enrolled by many students.

- These two ways analysis aims to formulate the "Bi-Directional relationship on data objects"

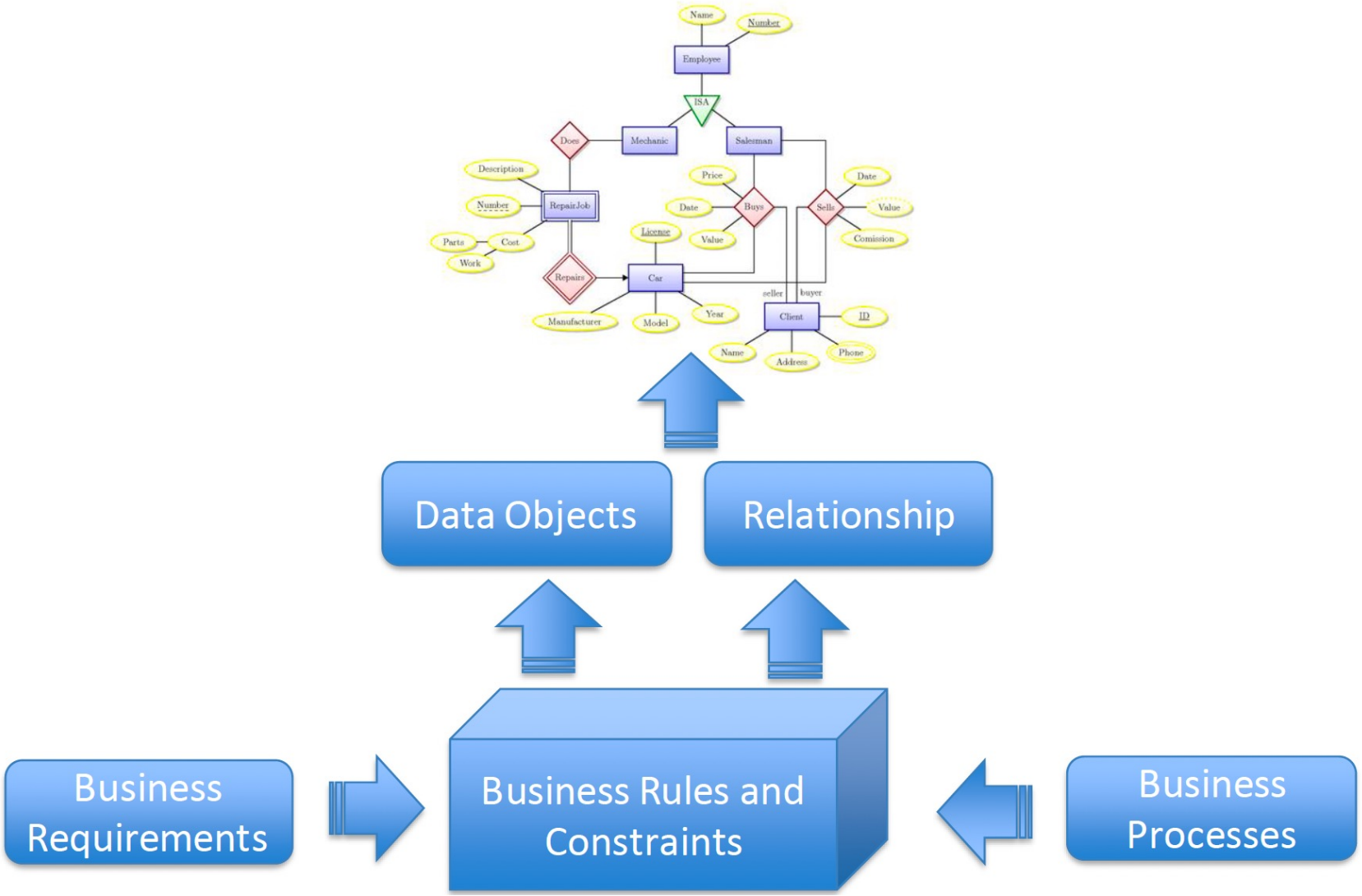


NOTE: This analysis can apply for TEMPLATE #1 only.

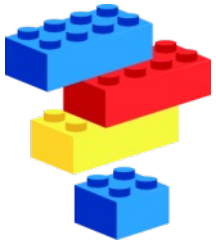
Business Constraints

- Represent the conditions or restrictions of business data observed from business activities
- For example,
  - A student must enroll for 5 courses, if the accumulated GPA is below 2.00.
  - A borrower must return the books within 7 days.
  - A taxi driver can rent a car for 7 days, but he can pay the rental fee for only 6 days.

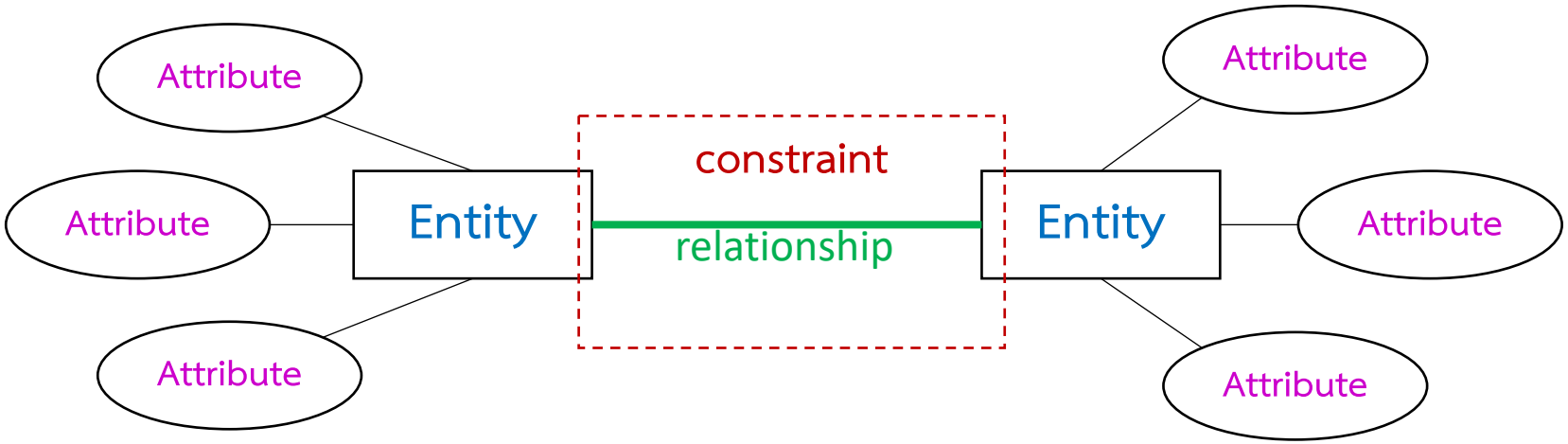
Business Rules as building box for data model creation



Data Model: Basic Building Blocks



- **Data Object** (defined as **Entity**) – anything about which data are to be collected and sorted.
- **Attribute** – characteristic or property of an entity.
- **Relationship** – association among entities.
- **Constraint** – restriction placed on the data.



Entity

- Data object (a person, a place, a thing, or an event) about which data are to be collected and stored.
- Represents a particular type of data object in the real world.
- They're "distinguishable" data object.
  - Each entity occurrence is unique and distinct.
- Maybe tangible or physical objects (for example, customers or products) OR intangible or abstractions (for example, flight routes or musical concerts)

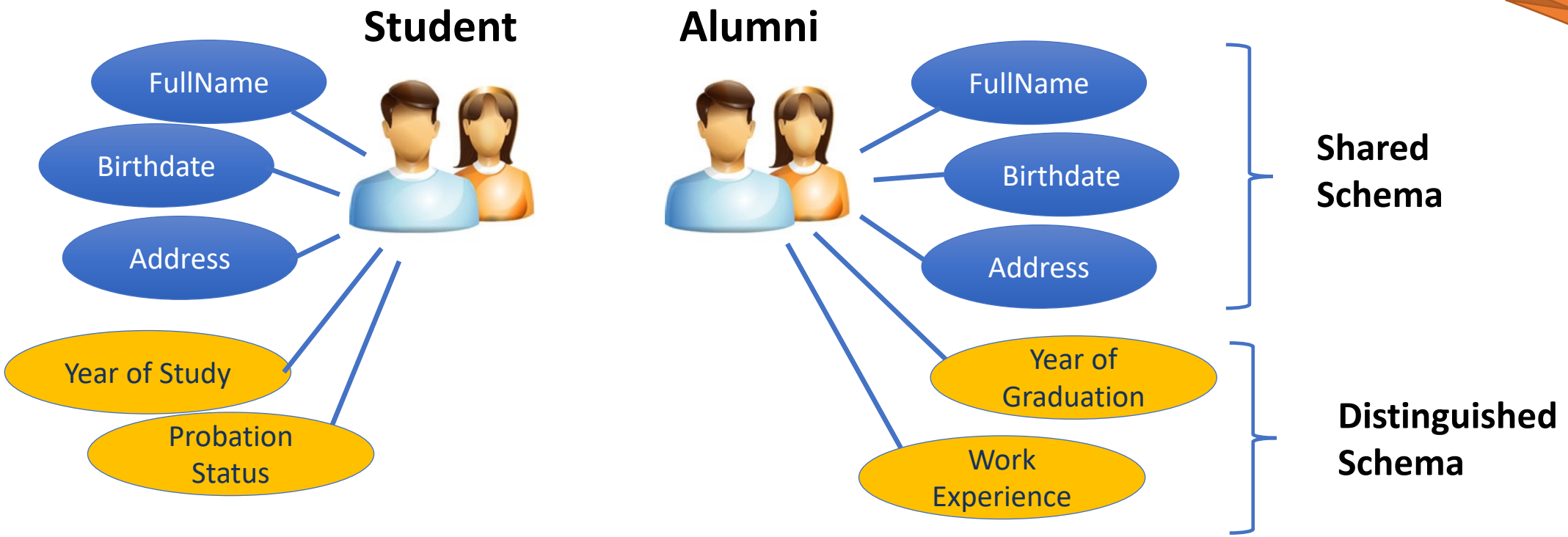
Distinguishable Data Object



- A "distinguishable data object" refers to a data object that can be uniquely identified and differentiated from other data objects within a given context.
- How to distinguish these data objects?
    - Student vs Alumni
    - Truck vs Bus
    - Customer vs Member

Distinguishable Data Object

Recommend



How about Truck vs Bus (and Customer vs Member)?

Master vs Transactional Data

Recommend

- Master data object is referred to the data obtained from the main data object. For example, student, customer, product, or department.
- Transactional data object relates to the transactions produced from business functions (or processes) and included data that is captured, for example, enrollment, order, payment, or reservation.

Product master data

Product ID	Product name	Product description	Price
1982SP	Lamp	Metal floor-standing lamp	56.95
454YH	Chair	Oak dining chair	70.99

Customer master data

Customer ID	Customer name	Customer address
67	J. Smith	56 Quy Road, Chester, UK
68	R. Hurst	14 Back Lane, Norwich, UK

Transactions

Sale ID	Product ID	Customer ID	Actual Sale price	Sale time
002	1982SP	67	56.95	1/3/13 15.34.12
003	454YH	67	65.99	1/3/13 15.34.25
004	454YH	68	70.99	4/3/13 12.05.43

Attribute

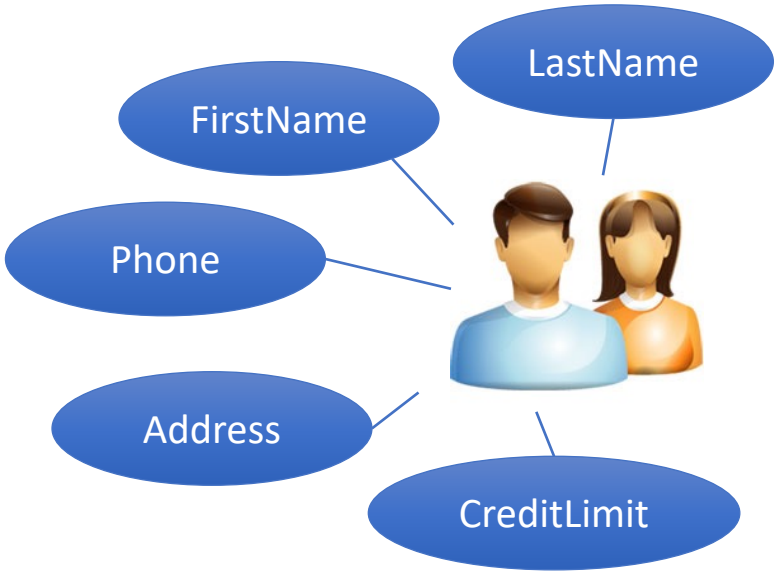
- A characteristic or property of an entity

Example:

- **CUSTOMER** entity is described by attributes:
  - Customer last name
  - Customer first name
  - Customer phone
  - Customer address
  - Customer credit limit.



How about CAR entity?



Relationship

- An association among entities.

FIGURE 2.1 LINKING RELATIONAL TABLES

Table name: AGENT (first six attributes)

Database name: Ch02_InsureCo

AGENT_CODE	AGENT_LNAME	AGENT_FNAME	AGENT_INITIAL	AGENT_AREACODE	AGENT_PHONE
501	Alby	Alex	B	713	228-1249
502	Hahn	Leah	F	615	882-1244
503	Okon	John	T	615	123-5589

Link through AGENT_CODE

Table name: CUSTOMER

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_AREACODE	CUS_PHONE	CUS_INSURE_TYPE	CUS_INSURE_AMT	CUS_RENEW_DATE	AGENT_CODE
10010	Ramas	Alfred	A	615	844-2573	T1	100.00	05-Apr-2016	502
10011	Dunne	Leona	K	713	894-1238	T1	250.00	16-Jun-2016	501
10012	Smith	Kathy	W	615	894-2285	S2	150.00	29-Jan-2017	502
10013	Olowski	Paul	F	615	894-2180	S1	300.00	14-Oct-2016	502
10014	Orlando	Myron		615	222-1672	T1	100.00	28-Dec-2017	501
10015	O'Brian	Amy	B	713	442-3381	T2	850.00	22-Sep-2016	503
10016	Brown	James	G	615	297-1228	S1	120.00	25-Mar-2017	502
10017	Williams	George		615	290-2556	S1	250.00	17-Jul-2016	503
10018	Farriss	Anne	G	713	382-7185	T2	100.00	03-Dec-2016	501
10019	Smith	Olette	K	615	297-3809	S2	500.00	14-Mar-2017	503

Example:

a relationship between CUSTOMER and AGENT:

An AGENT can serve many CUSTOMER

Each CUSTOMER may be served by one AGENT

Three Types of Relationship



One-to-many (1:M or 1..*) relationship

- A painter paints many different paintings, but each one of them is painted by only one painter

Many-to-many (M:N or *..*) relationship

- An employee may learn many job skills, and each job skill may be learned by many employees.

One-to-one (1:1 or 1..1) relationship

- Each of the stores be managed by a single employee.
- Each store manager, who is an employee, manages only a single store.

Note: Relationships are BIDIRECTIONAL ← →

How to determine the type of relationship?



Forward Sentence: A student can hold only one student card (1:1).

Backward Sentence: A student card can be held by only one student (1:1).

Analysis: **Forward (1:1) vs Backward (1:1) → Final conclusion is (1:1)**

Forward Sentence: A student can apply for only one faculty (1:1).

Backward Sentence: A faculty can be applied by many students (1:M).

Analysis: **Forward (1:1) vs Backward (1:M) → Final conclusion is (1:M)**

Forward Sentence: A student can enroll many subjects (1:M).

Backward Sentence: A subject can be enrolled by many students (1:M).

Analysis: **Forward (1:M) vs Backward (1:M) → Final conclusion is (M:N)**

Constraints

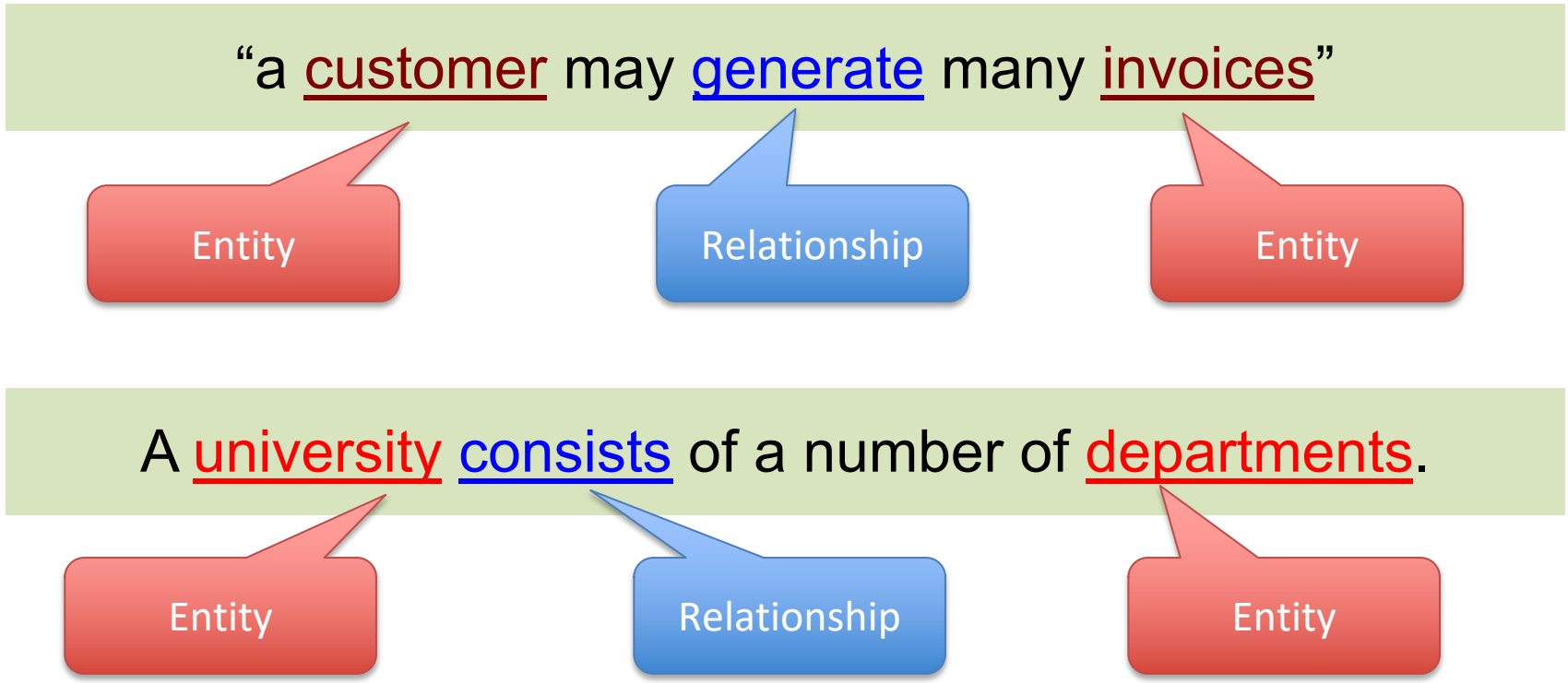
- A restriction placed on the data
- Ensure data integrity
- Normally expressed in the form of rules

- **Examples**:
  - An employee's salary must have values that are between 6,000 and 350,000.
  - A student's GPA must be between 0.00 and 4.00.
  - Each class must have one and only one teacher.

Example of Business Rules

A university consists of a number of departments. Each department offers several courses. A number of modules make up each course. Students enroll in a particular course and take modules towards the completion of that course. Each module is taught by a lecturer from the appropriate department, and each lecturer tutors a group of students.

Transforming Business Rules to Data Model (External/Conceptual mapping)



NOTE: Use the **template #1** from slide 20

Transforming Business Rules to Data Model (External/Conceptual Mapping)

- A statement of Business Rule provide:
  - Noun and
  - Verb

A noun ⇨ an **Entity** in the data model
A verb ⇨ a **Relationship** among the entities

- Generally, nouns translate to an Entity,
- Verbs translate to be a relationships among entities
- Relationship are bi-directional

More to Read

- Chapter 2: Database Systems from "***Database Systems: Design, Implementation, & Management***" **13th Edition** by Carlos Coronel and Steven Morris